

Assinment based on **Chapter 3: Software Quality Factors (McCall's Quality Model)**,

You will get a long gap for preparation. So, align yourselves for answering each. Part A submission will be appreciated if you can submit by next friday. Rest you will get another week for final submissions.

Part A: Short Questions (2–3 marks each)

1. Why are software quality requirements considered non-functional requirements?
2. What is the difference between functional requirements and quality factors?
3. List the three categories of McCall's Software Quality Factors.
4. Define **Correctness** with an example.
5. What is meant by **Reliability**? Give one measurable reliability requirement.
6. Explain the role of **Integrity** in software quality.
7. What is the difference between **Maintainability** and **Flexibility**?
8. Why is **Testability** important during software development?
9. Define **Portability** with a practical example.
10. What is **Interoperability** and why is it important in enterprise software?

Part B: Analytical Questions (5 marks each)

1. Compare the three categories of McCall's Software Quality Factors using suitable examples.
2. Explain how poor software requirements can negatively affect software quality.
3. A hospital information system requires 99.999% availability. Which McCall quality factors are most critical? Justify your answer.
4. Distinguish between **Reliability**, **Safety**, and **Survivability**.
5. Explain why **Reusability** can reduce software development cost and time.

Part C: Scenario-Based Questions (8–10 marks each)

1. A banking application must:
 - Prevent unauthorized access,
 - Process transactions within 2 seconds,
 - Operate continuously with minimal downtime.
2. Identify the relevant software quality factors and explain why each is important.

3. An e-commerce company wants its application to run on Windows, Linux, Android, and cloud servers while integrating with several payment gateways. Which McCall quality factors should receive the highest priority? Justify your answer.
4. A software company frequently modifies its product for different clients. Which Product Revision Factors are most important? Explain with examples.
5. Compare McCall's model with the alternative models proposed by Evans & Marciniak and Deutsch & Willis. Which additional quality factor do you consider most valuable in modern software engineering? Explain.

Part D: Higher-Order Thinking (10–15 marks each)

1. Critically evaluate whether McCall's Software Quality Model is still relevant for modern cloud-native and AI-based software systems.
2. Design a quality requirements specification for an online university learning management system using McCall's quality factors.
3. Explain how software quality factors influence Software Requirements Specification (SRS). Discuss with examples.
4. If you were designing an autonomous vehicle control system, which quality factors would you prioritize and why?

Viva/Discussion Questions

- Which software quality factor is the most difficult to measure? Why?
- Can software be highly efficient but have poor usability? Explain.
- Why should clients be involved in defining software quality requirements?
- Are quality factors equally important for every software project? Justify your answer.
- Which quality factor is most critical for AI-powered healthcare systems?